Stat 375 Inference in Graphical Models

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This notes provide basic definitions of factor graphs, pairwise graphical models and introduce an important inference algorithm: belief propagation. As a motivational example, we showed in class that a diagnostic-type Bayesian network can be reduced (in an appropriate sense) to a factor graph. This example can be found in [MM09, Section 13.1].

1 Factor graphs

The first subsection introduces factor graphs, and the second shows how general factor graphs are equivalent to pairwise models.

1.1 Basic definitions

A factor graph is a bipartite graph $G = (V, F, E)$, whereby V and F are two (finite) sets of vertices, and $E \subseteq V \times F$ a set of undirected edges. We will often identify $V = [n]$ (set of the first n integers), $F = [m]$. We will call *variable nodes* the vertices in V, and use for them letters $i, j, k, \dots$, and *function* (or *factor*) *nodes* the vertices in F to be denoted by $a, b, c \dots$.

Given $i \in V$, the set of its neighbors is denoted by $\partial i = \{a \in F : \ (i, a) \in E\}$. The neighborhood of $a \in F$, denoted by ∂a, is defined analogously.

A *factor graph model* (or *graphical model*) specifies the joint distribution of random variables $\underline{X} = (X_1, \dots, X_n) = \{X_i : i \in V\}$ taking value in a domain $\mathcal{X}$. We shall generally assume[1] that either $\mathcal{X}$ is a finite set, or $\mathcal{X} = \mathbb{R}$. Indeed we shall mostly focus on the first case and write general equations in discrete notation. Finally, for any subset of variable nodes $A \subseteq V$, we write $\underline{x}_A = \{x_i : i \in A\}$.

A specific model is defined by the triple $(G, \mathcal{X}, \underline{\psi})$ where $\underline{\psi} = (\psi_1, \dots, \psi_m) = \{\psi_a : a \in F\}$ is a vector of functions $\psi_a : \mathcal{X}^{\partial a} \to \mathbb{R}_+$. The ψ_a's are referred to as *potentials* or *compatibility functions*. The joint distribution $\mathbb{P}\{\underline{X} = \underline{x}\} \equiv \mu(\underline{x})$ takes the form

$$\mu(\underline{x}) = \frac{1}{Z} \prod_{a \in F} \psi_a(\underline{x}_{\partial a}) \,, \tag{1}$$

where the normalization

$$Z \equiv \sum_{\underline{x}} \prod_{a \in F} \psi_a(\underline{x}_{\partial a}) \,, \tag{2}$$

is called the *partition function*. Notice that, in order for the distribution (1) to be well defined, there must be at least one configuration $\underline{x} \in \mathcal{X}^V$ such that $\psi_a(\underline{x}_{\partial a}) > 0$ for all $a \in F$.

In the case $\mathcal{X} = \mathbb{R}$, Eq. (1) defines the density of $(X_1, \dots, X_n)$ with respect to the product Lebesgue measure (the ψ_a's have to be measurable in this case).

1.2 Pairwise graphical models

Pairwise graphical models, also called Markov random fields (MRFs) are defined in terms of a simple graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$. It is convenient to introduce a compatibility function $\psi_i : \mathcal{X} \to$

[1] In principle, the formalism can be defined whenever $\mathcal{X}$ is a measure space, but I do not know of any example that is worth the effort.

$\mathbb{R}_+$ for each vertex $i \in \mathcal{V}$, and one $\psi_{ij} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_+$ for each edge $(i,j) \in \mathcal{E}$. The joint distribution of $(X_1, \dots, X_n)$, $\mathbb{P}(\underline{X} = \underline{x}) = \mu(\underline{x})$ is then defined by

$$\mu(\underline{x}) = \frac{1}{Z} \prod_{(i,j)\in\mathcal{E}} \psi_{ij}(x_i, x_j) \prod_{i\in\mathcal{V}} \psi_i(x_i) \tag{3}$$

It is important to remember that pairwise models are equivalent to factor graph models: neither family is more powerful than the other. Choosing which one to use is a matter of convenience.

In order to check this statement one has to construct, for any pairwise model $\mathcal{M}$, an associated factor graph model M which describes the same distribution $\mu(\,\cdot\,)$, and viceversa. Reduction from pairwise models to factor graphs is straightforward. Use $\mathcal{V}$ as the set of variable nodes, and associate to each edge $(i,j) \in \mathcal{E}$ and to each vertex $\mathcal{V}$ a factor node (in other words, set $V = \mathcal{V}$ and $F \simeq \mathcal{V} \cup \mathcal{E}$).

Reduction from factor graphs to pairwise models is slightly less trivial. The basic idea is to replace each factor node a by an ordinary vertex and associate to it a variable $x_a \in \mathcal{X}^{\partial a}$ which keeps track of the values $\{x_i :\in \partial a\}$. We will fill the details in class.

2 Belief propagation

Belief propagation (BP) is a term describing a family algorithms for approximate inference in graphical models. An introduction to BP can be found in [MM09, Chapter 14]. Here we limit ourselves to a summary of equations that might be helpful for future reference. Thrughout this course, the symbol $\cong$ is used to denote equality between functions up to a multiplicative normalization. Formally, we shall write $f(x) \cong g(x)$ if there exist a non-vanishing constant A such that $f(x) = Ag(x)$ for all x.

3 Sum-product

Under parallel update scheduling we have:

$$\nu^{(t+1)}_{j\to a}(x_j) \;\cong\; \prod_{b\in\partial j\setminus a} \widehat{\nu}^{(t)}_{b\to j}(x_j)\,, \tag{4}$$

$$\widehat{\nu}^{(t)}_{a\to j}(x_j) \;\cong\; \sum_{\underline{x}_{\partial a\setminus j}} \psi_a(\underline{x}_{\partial a}) \prod_{k\in\partial a\setminus j} \nu^{(t)}_{k\to a}(x_k)\;. \tag{5}$$

It is understood that, when $\partial j \setminus a$ is an empty set, $\nu_{j\to a}(x_j)$ is the uniform distribution.

After t iterations, one can estimate the marginal distribution $\mu(x_i)$ of variable i using the set of *all* incoming messages. The BP estimate is:

$$\nu^{(t)}_i(x_i) \cong \prod_{a\in\partial i} \widehat{\nu}^{(t-1)}_{a\to i}(x_i)\,. \tag{6}$$

4 Max-product

Max-product updates are used to approximately find the mode of a distribution specified by a gactor graph model. They read

$$\nu^{(t+1)}_{i\to a}(x_i) \;\cong\; \prod_{b\in\partial i\setminus a} \widehat{\nu}^{(t)}_{b\to i}(x_i)\,, \tag{7}$$

$$\widehat{\nu}^{(t)}_{a\to i}(x_i) \;\cong\; \max_{\underline{x}_{\partial a\setminus i}} \left\{ \psi_a(\underline{x}_{\partial a}) \prod_{j\in\partial a\setminus i} \nu^{(t)}_{j\to a}(x_j) \right\}\;. \tag{8}$$

Equation (6) can be used to approximate the *max-marginals* defined by

$$M_i(x_i^*) = \max_{\underline{x}} \{\mu(\underline{x}) : x_i = x_i^*\} \ . \tag{9}$$

These updates can be rewritten also in the min-sum form. The problem is in this case set up as the one of minimizing a cost function that factorizes according to G:

$$H(\underline{x}) = \sum_{a \in F} H_a(\underline{x}_{\partial a}) \, . \tag{10}$$

We refer to [MM09, Section 14.3] for their explicit form (It is sufficient to take the logarthm of Eqs. (7) and (7).)

References

[MM09] M. Mézard and A. Montanari. *Information, Physics and Computation.* Oxford University Press, Oxford, 2009. Available online: `http://www.stanford.edu/ montanar/BOOK`.